

EcoPassEU: A Privacy-Aware Digital Product Passport DApp for Sustainable Consumer Goods in Europe

Fuyu Cao¹, Luxiao Cao¹, Detai Dong¹, Akshansh Rajora¹, and Tianwei Yu¹

University of Bristol, Bristol, UK

{1d25515, ne25392, mz25325, tk25240, np25024}@bristol.ac.uk

1 Introduction

Every year, billions of products enter the European market with sustainability labels, yet consumers have little reliable way of checking whether claims such as “low carbon” or “ethically sourced” are accurate. A study commissioned by the European Commission found that more than half of the green environmental declarations lack verifiable evidence to support, eroding public confidence in corporate environmental commitments [16].

To address this credibility gap, the Ecodesign for Sustainable Products Regulation (ESPR) now mandates Digital Product Passports (DPPs) —standardised lifecycle records—for most EU products [7]. With the implementation of compliance in 2027, reliable DPP solutions have become a market urgent need[7].

Our product EcoPassEU is a blockchain-based DPP platform which allows manufacturers to record product information as permanent, publicly verifiable entries on the Ethereum blockchain. Unlike traditional databases, the immutability of blockchain records means that once the data is released, it cannot be altered retrospectively, eliminating the possibility of post-hoc falsification [5]. Any user can obtain a complete and encrypted product history by entering the product ID or scanning the QR code.

EcoPassEU serves four key stakeholders. It provides SME manufacturers with a cost-effective ESPR compliance channel through a web for batch data entry and automated QR code generation. At the same time, consumers, regulators and retailers can also get open and wallet-free access. Just search for the product ID to immediately obtain the blockchain-verified product passport and supply chain map, enabling the independent verification of environmental claims without prior technical expertise.

The value of EcoPassEU is threefold. Primarily, it provides manufacturers with a practical way to ESPR deadlines, reducing the risk of market exclusion. Meanwhile, the platform directly cracked down on greenwashing, securing product records via cryptographic hashing replaces brand self-reporting with independent, third-party verification. Thirdly, it promotes the development of circular economy by equipping recyclers with the precise material data necessary to recover resources when products are scrapped.

2 Background

2.1 Digital Product Passports and the Regulatory Context

A Digital Product Passport (DPP) is a structured digital record encompassing lifecycle data—such as material composition, provenance, and end-of-life guidance—designed to inform stakeholders and support more sustainable design choices, efficient resource use, and the transition towards circular business models [1,2,4,12].

The regulatory foundation for DPPs comes from the European Green Deal and Circular Economy Action Plan [10,11]. Based on these 2 frameworks, the Ecodesign for Sustainable Products Regulation (ESPR), came into force on 18 July 2024,, mandates DPPs for almost all physical goods sold in the EU, regardless of the seller’s location [7].

2.2 The Problem: Trust in Sustainability Claims

Despite regulatory momentum, the implementations of most DPPs rely on centralised, proprietary databases [15]. This architecture encourages the selective concealment of data, hindering aftermarket participants like independent repairers [8]. Furthermore, managing huge volume of lifecycle data centrally raises significant concerns regarding storage efficiency, data integrity, and vulnerability to breaches[19]. Taken together, downstream actors — including importers, recyclers, and end consumers — lack reliable ways to independently verify product information.

This problem is exacerbated by the prevalence of greenwashing, broadly understood as false, vague, or unsubstantiated environmental claims that make products appear more sustainable than they are [13]. A 2020–2021 EU sweep of 344 online claims found that nearly half were potentially deceptive—often substituting verifiable details with vague terms like “conscious”—and 59% lacked accessible evidence [16]. Another EU website screening found that 42% of sustainability claims were misleading enough to constitute improper commercial behaviour [16].

Blockchain provides a structural solution to this trust problem. Because records on a public blockchain are replicated across a distributed network, neither single party can unilaterally tamper with the records, providing a form of tamper evidence that cannot be replicated by a centralised system [5,14]. Smart contract introduced in the Ethereum protocol allows data access and update rules to be encoded in publicly readable logic, so that there is no need for trusted intermediaries and enabling cross-organisational verification without shared infrastructure [21].

2.3 Related Work and Existing Platforms

Blockchain-based product traceability uses its immutability and decentralised verification to improve supply chain transparency [5]. Blockchain-based product traceability emerged as a research topic around 2015–2016, with early work

concentrated in food and agriculture before spreading to pharmaceuticals, electronics, and textiles [17,9]. More recently, research has shifted from feasibility towards scale and efficiency: for instance, Walmart’s use of IBM Food Trust — built on Hyperledger Fabric — to reduce food-item tracing time from six days to under three seconds [9] .

In the commercial field, Circularise is among the closest comparable platforms, offering DPPs with selective disclosure of sustainability data across supply chains, with applications spanning the automotive, battery, and textile sectors [6]. VeChain similarly deploys blockchain and smart contracts on its VeChainThor platform to achieve supply chain transparency, though it runs on a permissioned network with a restricted set of authority-approved validators [18] Both platforms have a common limitation: reliance on proprietary infrastructure or permissioned networks introduces vendor dependency and reduces interoperability [3,20].

2.4 Business Viability

3 Design Documentation

Outline the system architecture of EcoPassEU. Include your justifications for using EVM-compatible networks like Polygon, ERC-1155 tokens, IPFS, and DIDs/VCs. Insert your Use Case diagrams, Sequence/Flow diagrams, and front-end wireframes here.

4 Project Execution

Detail the implementation process of the DApp. This should feature core code analysis, data structures, your testing strategy, final UI screenshots, and a brief user manual on how to interact with the application.

4.1 Critical Evaluation

Provide a technical analysis of EcoPassEU, evaluating the gas usage of your smart contracts, security considerations (especially regarding the separation of on-chain and off-chain data), and runtime performance.

5 Conclusion

Summarize the main theoretical and technical findings. Detail what the group achieved, acknowledge limitations (e.g., the physical-to-digital bridge vulnerabilities), and propose future work.

References

1. Adisorn, T., Tholen, L., Götz, T.: Towards a digital product passport fit for contributing to a circular economy. *Energies* **14**, 2289 (2021). <https://doi.org/10.3390/en14082289>
2. Basal, M., Demircioglu, A.: Digital product passport in marketing and the future of sustainable development. *American Journal of Industrial and Business Management* (2024). <https://doi.org/10.4236/ajibm.2024.145039>
3. Belchior, R., Vasconcelos, A., Guerreiro, S., Correia, M.: A survey on blockchain interoperability: Past, present, and future trends. *ACM Computing Surveys (CSUR)* **54**, 1 – 41 (2020). <https://doi.org/10.1145/3471140>
4. van Capelleveen, G., Vegter, D., Olthaar, M., van Hillegersberg, J.: The anatomy of a passport for the circular economy: a conceptual definition, vision and structured literature review. *Resources, Conservation & Recycling Advances* (2023). <https://doi.org/10.1016/j.rcradv.2023.200131>
5. Casino, F., Dasaklis, T.K., Patsakis, C.: A systematic literature review of blockchain-based applications: Current status, classification and open issues. *Telematics and Informatics* **36**, 55–81 (2019). <https://doi.org/10.1016/j.tele.2018.11.006>
6. Circularise: Circularise: Product traceability platform for supply chain compliance. <https://www.circularise.com/> (2025), accessed: April 2026
7. Circularise: A guide to the ecodesign for sustainable products regulation (ESPR). <https://www.circularise.com/blogs/a-guide-to-the-ecodesign-for-sustainable-products-regulation-espr> (2025), accessed: April 2026
8. Ducuing, C., Reich, R.: Data governance: Digital product passports as a case study. *Competition and Regulation in Network Industries* **24**, 3 – 23 (2023). <https://doi.org/10.1177/17835917231152799>
9. Ellahi, R.M., Wood, L.C., Bekhit, A.: Blockchain-based frameworks for food traceability: A systematic review. *Foods* **12** (2023). <https://doi.org/10.3390/foods12163026>
10. European Commission: The european green deal. https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal_en (2019), accessed: 28 April 2026
11. European Commission: A new circular economy action plan: For a cleaner and more competitive europe. Communication from the Commission COM(2020) 98 final, European Commission, Brussels (2020)
12. Jansen, M., Meisen, T., Plociennik, C., Berg, H., Pomp, A., Windholz, W.: Stop guessing in the dark: Identified requirements for digital product passport systems. *Syst.* **11**, 123 (2023). <https://doi.org/10.3390/systems11030123>
13. Kundi, V., Ernszt, I.: The phenomenon of greenwashing: An analysis of the hungarian regulation. *Journal of Sustainability Research* (2024). <https://doi.org/10.20900/jsr20240066>
14. Nakamoto, S.: Bitcoin: A peer-to-peer electronic cash system. <https://bitcoin.org/bitcoin.pdf> (2008), accessed: April 2026
15. Psarommatis, F., May, G.: Digital product passport: A pathway to circularity and sustainability in modern manufacturing. *Sustainability* (2024). <https://doi.org/10.3390/su16010396>
16. Silva, D.: The fight against greenwashing in the european union. *UNIO – EU Law Journal* (2021). <https://doi.org/10.21814/unio.7.2.4029>

17. Tian, F.: An agri-food supply chain traceability system for China based on RFID & blockchain technology. In: 2016 13th International Conference on Service Systems and Service Management (ICSSSM). IEEE (2016). <https://doi.org/10.1109/ICSSSM.2016.7538424>
18. VeChain: Vechain official: Real-world impact blockchain technology. <https://vechain.org/> (2026), accessed: April 2026
19. Voulgaridis, K., Lagkas, T.D., Angelopoulos, C., Boulogeorgos, A.A., Argyriou, V., Sarigiannidis, P.G.: Digital product passports as enablers of digital circular economy: a framework based on technological perspective. *Telecommunication Systems* **85**, 699–715 (2024). <https://doi.org/10.1007/s11235-024-01104-x>
20. Wang, G., Wang, Q., Chen, S.: Exploring blockchains interoperability: A systematic survey. *ACM Computing Surveys* **55**, 1 – 38 (2023). <https://doi.org/10.1145/3582882>
21. Wood, G.: Ethereum: A secure decentralised generalised transaction ledger. Tech. rep., Ethereum Foundation (2014), yellow Paper. Available at <https://ethereum.github.io/yellowpaper/paper.pdf>

A Appendix

This appendix includes the mandatory materials required for the FTGP summative assessment.

A.1 Smart Contract Deployment

- **Smart Contract Address(es):** [Insert Sepolia testnet addresses here]
- **Etherscan Links:** [Insert links to verified contracts on Etherscan]

A.2 Project Links

- **Deployed Website:** [Insert URL]
- **GitHub Repository:** https://github.com/DeTaiDong/FTGP2425_Group1_11

A.3 Transaction Hashes

- **Deployment TX:** [Insert Hash] - Explanation.

A.4 Equity Shares & Contributions

- Luxiao Cao: [Weight]
- Akshansh Rajora: [Weight]
- Fuyu Cao: [Weight]
- Tianwei Yu: [Weight]
- Detai Dong: [Weight]

[Insert Screenshot of GitHub - Insights - Contributor here]

A.5 JIRA Project Board

[Insert screenshots of 'Summary', 'timeline', and 'backlog' pages here]

A.6 Individual Reflection Essays

Luxiao Cao [Insert 500-word reflection here]

Akshansh Rajora [Insert 500-word reflection here]

Fuyu Cao [Insert 500-word reflection here]

Tianwei Yu [Insert 500-word reflection here]

Detai Dong [Insert 500-word reflection here]

A.7 Team Sprint Reports

[Insert chronological sprint reports here]

A.8 Complete Source Code

Smart Contracts

```
// Insert complete Solidity code here
```

Frontend Core Files

```
// Insert modified/relevant frontend files (e.g., page.js) here
```